EcoPulse

Citizen Carbon Accounting & Rooftop Solar Adoption

A mobile-first platform operationalizing India's **Mission LiFE** through deterministic emission auditing, micro-nudges, and verified PM Surya Ghar rooftop solar planning.

Track: HC26-GEDA01 (Gujarat Energy Development Agency)

Team: [TEAM_NAME]

Members: [MEMBERS]

Code Repository: [REPO_URL]

Live Web Demo: [DEMO_URL]

Video Pitch: [VIDEO_URL]

The Problem

The Gap Between Climate Awareness & Daily Action

Citizens recognize climate urgency but lack personalized, transparent tools to quantify domestic emissions, evaluate solar feasibility, and sustain verifiable lifestyle changes.

Urban Residents

Unaware of specific electricity and fuel footprints; overwhelmed by complex rooftop solar policies and subsidy paperwork.

Local Communities

Zero localized emissions benchmarks, peer comparison, or collective mechanisms to celebrate neighbourhood climate leadership.

State Administrators

Lack verifiable, bottom-up citizen adoption data across Gujarat districts to monitor Mission LiFE targets effectively.

Focus: Track HC26-GEDA01 — Empowering Gujarat households with transparent sustainability tools.

The Solution

Four Core Capabilities for Citizen Climate Action

EcoPulse bridges awareness and impact through verifiable math, habit reinforcement, and end-to-end solar onboarding.

1. Footprint Tracker

BUILT

Deterministic calculator covering electricity, cooking gas (LPG/PNG), and multi-modal transport with local Gujarat benchmarks.

2. Action Center & Micro-Nudges

BUILT

14 categorized Mission LiFE habits (Energy, Water, Waste, Mobility, Food) with daily check-ins and append-only points ledger.

3. PM Surya Ghar Solar Hub

BUILT

3-step solar ROI wizard, DISCOM tariff inversion, official subsidy breakdown, and verified Gujarat installer directory.

4. Community & Admin View

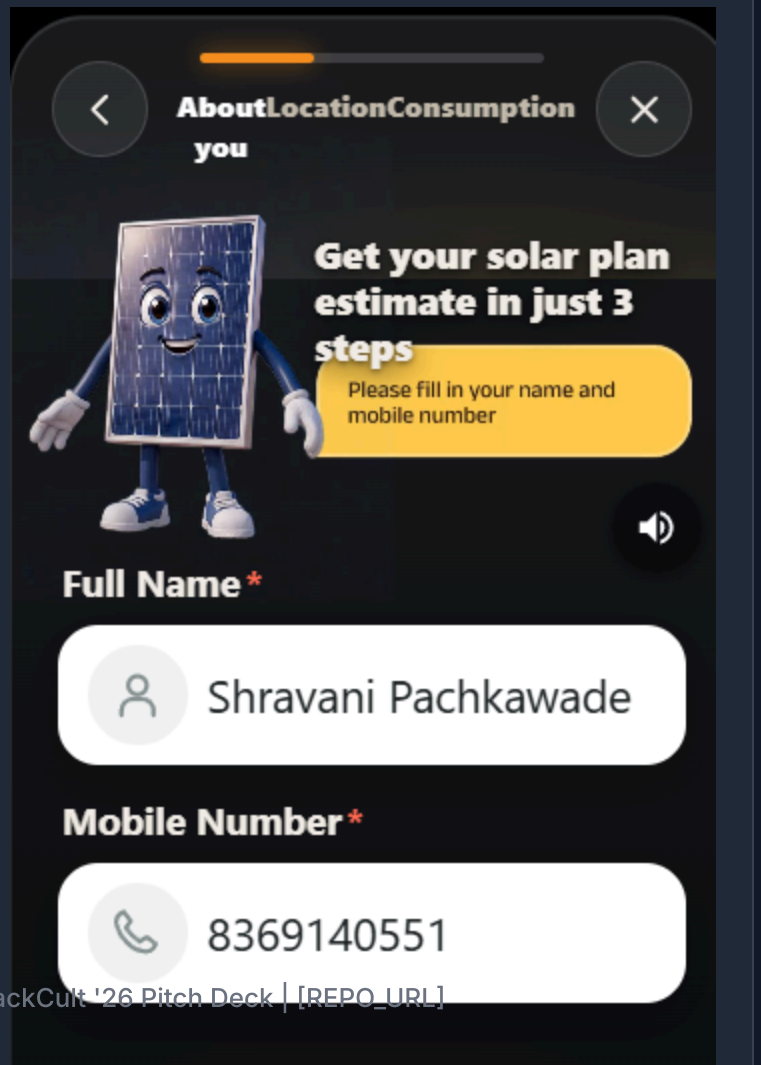
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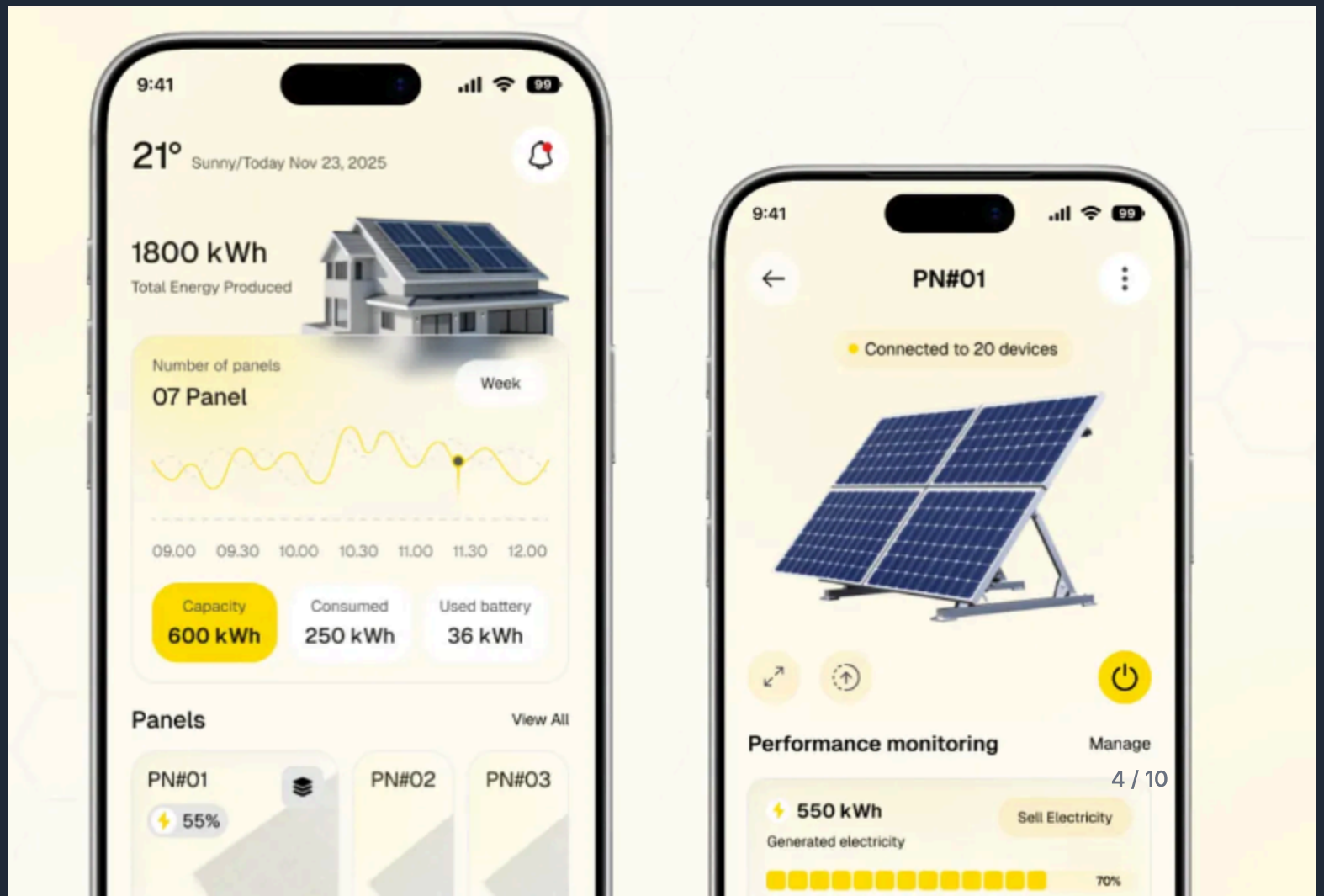
Society leaderboards and GEDA administrative analytics for district solar adoption tracking (schema & transactions built).

Product Walkthrough

Mobile-First, Tri-Lingual & Verifiable Interface

Designed for high accessibility across Gujarat with instant preview, audio guidance, and clear status badges.

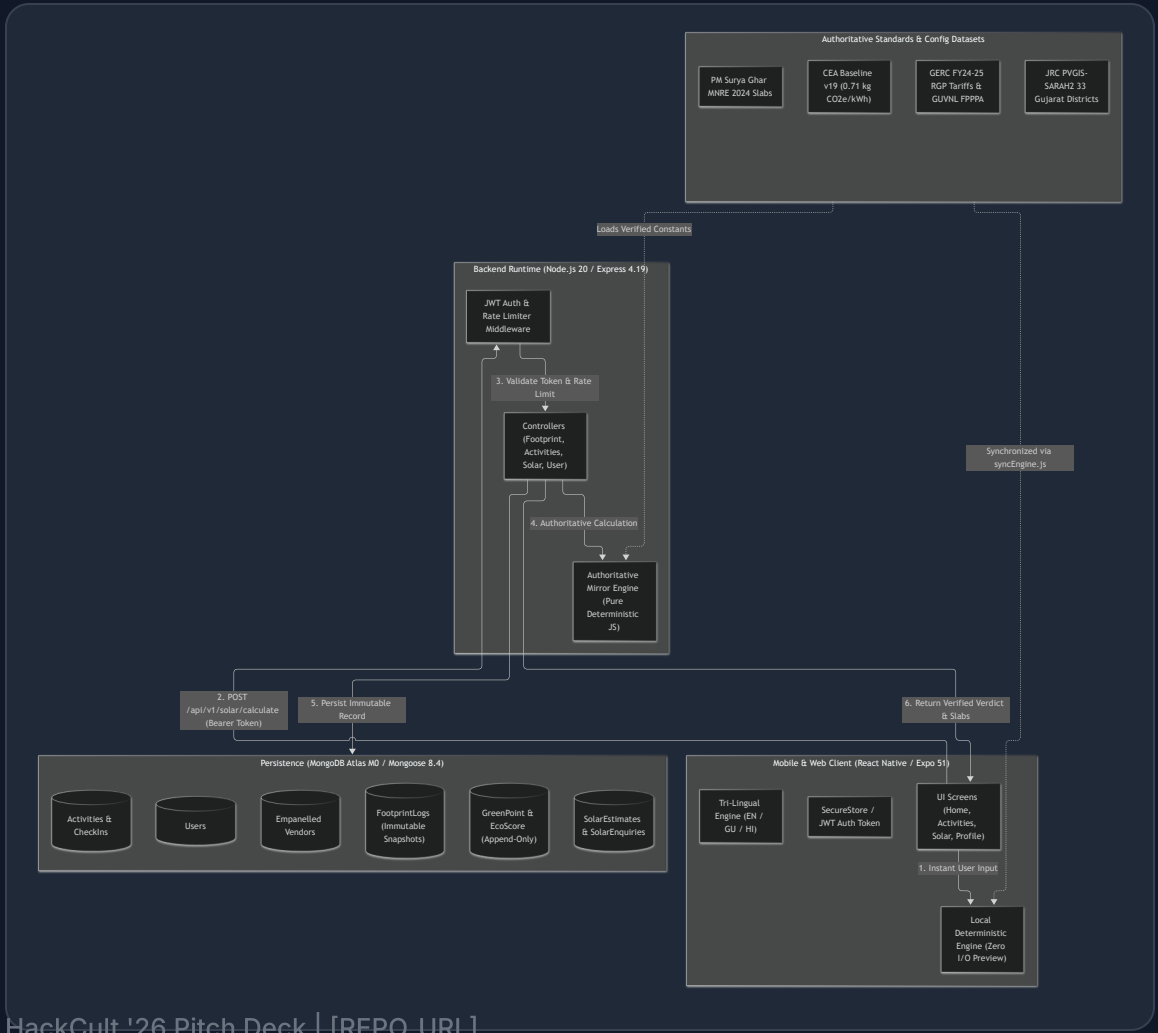




System Architecture

Mirrored Deterministic Engine & Resilient API

Dual-engine architecture delivers zero-latency client calculation while enforcing authoritative server-side persistence.



Numbered Request Lifecycle

1. **Client Preview:** Local pure engine computes instant preview with zero network round-trip.
2. **Authenticated Request:** Client dispatches payload with JWT Bearer token to Express REST API.
3. **Security & Throttle:** Auth middleware verifies token; rate-limiter prevents denial-of-service.
4. **Authoritative Calculation:** Server recomputes metrics using identical synced engine code.
5. **Immutable Persistence:** Snapshot saved to MongoDB Atlas (`FootprintLog` / `SolarEstimate`).
6. **Verified Response:** Authoritative verdict and subsidy breakdown returned to client.

Calculation Engine & Standards

Verifiable Climate Accounting Across 24/24 Automated Tests

Zero inline magic numbers; all factors are loaded from versioned datasets and verified against government guidelines.

Core Scientific Standards

- **Grid Emission Factor:** 0.71 kg CO_2e/kWh Verified
  CEA CO_2 Baseline Database Ver 19 (2024), Western Grid
- **Cooking Fuel:** LPG 2.98 kg/kg, PNG 2.15 kg/SCM Verified
  IPCC 2006 Guidelines & PPAC Benchmark
- **Rooftop Yield:** 33 Gujarat Districts Verified
  European Commission JRC PVGIS-SARAH2 API v5.2
- **DISCOM Tariffs:** UGVCL, DGVCL, MGVCL, PGVCL, Torrent Verified
  GERC Tariff Orders FY 2024-25 RGP & GUVNL FPPPA
- **Central Subsidy:** ₹30k/kW (1-2kW), ₹78k cap Verified
  MNRE PM Surya Ghar Operational Guidelines (2024)

Engine Integrity & Verification

Engine Sync Verification: `scripts/checkEngineSync.js` guarantees byte-for-byte identity between `shared/engine/` and `client/src/engine/`.

Automated Test Suite: 24/24 PASS

Validates subsidy slabs, 25-year degradation (0.5%/yr), bisection bill inversion, and district yield data.

Estimate **Capital Cost (₹50k-60k/kW) & State Top-up Subsidy pending GEDA tender discovery.**

Database Schema

Relational Integrity with Append-Only Audit Trails

MongoDB schemas enforce strict foreign-key referencing, compound index constraints, and immutable records.

Append-Only Points Ledger

`GreenPointTransaction` records are strictly append-only with compound unique index `{ activityCycleId: 1, transactionType

```
<div class="card">
  <h3 style="color: #10B981;">Immutable Baseline Snapshot</h3>
  <p style="font-size: 16px; margin: 0;">`FootprintLog` captures an immutable `rawInputsSnapshot` at onboard
</div>
```

Tech Stack

Selected from Repository Dependency Manifests

Modern, lightweight stack prioritizing offline reliability, high developer velocity, and unified JavaScript syntax.

Layer	Technology	Repository Source & Package Reason
Mobile Client	**React Native 0.88 / Expo 58**	`client/package.json` — Rapid cross-platform native compilation and sensor access
Web Preview	**React Native Web 0.21**	`client/package.json` — Zero-duplication browser rendering for evaluator access
Client Cache	**@tanstack/react-query 5.105**	`client/package.json` — Resilient query caching, auto-retry, and offline fallback
Backend API	**Node.js 20 / Express 4.19**	`server/package.json` — High-concurrency event-driven I/O with unified JS calculation
Database	**MongoDB Atlas / Mongoose 8.4**	`server/package.json` — Flexible nested survey schemas and multikey district indexing
Authentication	**JWT / bcryptjs 2.4**	`server/package.json` — Stateless, mobile-friendly authorization via bearer tokens
Verification	**Node.js Assert & Sync Scripts**	`tests/solar.test.js` — Zero-dependency deterministic verification suite

Value & Scalability

Tangible Citizen Impact with High-Throughput Architecture

Connecting personal utility savings with state-level decarbonization goals.

Stakeholder Value

- **Citizens:** ₹78,000 subsidy clarity, <6% green loans, and up to ₹9.87L lifetime electricity bill savings.
- **Communities:** Shared accountability for Mission LiFE pledges across housing societies.
- **GEDA (Planned):** Aggregated district-level solar adoption metrics and empanelled vendor load visibility.

Implemented Scalability (in Code)

- **Stateless API:** JWT bearer authentication allows horizontal Node.js clustering without session stickiness.
- **Offloaded Client Compute:** Dual pure engine computes previews on device, slashing backend CPU load.
- **Compound Indexing:** MongoDB indexes on `{ userId: 1, createdAt: -1 }` and `{ district: 1 }` ensure sub-10ms queries.

Scaling Path (Planned): Redis caching for static district solar yield matrices; read-replicas for administrative analytics; queue-backed push notifications via BullMQ.

Limitations & Roadmap

Transparent Engineering Constraints & Delivery Milestones

Grounded in actual repository code; milestones prioritize data authority and state-level rollouts.

Current Limitations (in Repo)

- **Self-Reported Inputs:** Relies on user bill inputs without direct DISCOM smart-meter API linkage.
- **Seasonal Normalization:** Baseline comparison lacks weather/temperature normalization.
- **Single Grid Factor:** Uses regional average (0.71 kg/kWh) rather than hourly marginal carbon intensity.
- **Cost Discovery:** Solar EPC pricing uses benchmark estimates pending official GEDA tender discovery.
- **Partial Translation:** English complete; Gujarati and Hindi localized in core UI keys.

Phased Engineering Roadmap

Next 2 Weeks (Polish):
Confirm GEDA state top-up subsidy; verify vendor EPC price bands; finalize 100% Gujarati translation strings.

Next 3 Months (Expansion):
Deliver GEDA Administrative Portal with district analytics; deploy society leaderboard challenges.

Next 12 Months (Integration):
Integrate GUVNL DISCOM automated bill sync and dynamic time-of-day grid emission signals.